THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is the best explanation for the cooling in a desert cooler? 1

(a) fan makes cold
(b) pads produce cold
(c) water is cold
(d) water evaporates and takes heat from the air

2. The best way to dry a wet handkerchief quickly is to: 1

(a) spread it open in warm, dry, windy place **(b)** dip it in cold water
(c) fold it and keep in a closed box **(d)** keep in a damp room

3. Assertion (A): Naphthalene balls disappear without leaving liquid. Reason (R): They undergo sublimation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A beaker of water is heated and bubbles appear at the bottom well before 100 degrees Celsius. They are mainly: 1

(a) steam **(b)** dissolved air coming out
(c) oxygen from water **(d)** hydrogen

5. Camphor kept in an open dish disappears but wax melts. This is because camphor: 1

(a) sublimes, wax melts then remains **(b)** freezes
(c) melts like wax **(d)** burns

6. Assertion (A): Water boils at a lower temperature on a mountain top. Reason (R): Atmospheric pressure is lower at the top. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. Naphthalene and ammonium chloride can both be separated from sand by: 1

(a) filtration **(b)** evaporation **(c)** decantation **(d)** sublimation

8. A gas is cooled until it turns into a liquid and then a solid. Compared with the gas, the solid has: 1

(a) no forces **(b)** particles far apart
(c) faster particles **(d)** particles close and in fixed positions

9. Assertion (A): The rate of diffusion is greater in gases than in liquids. Reason (R): Gas particles are free to move and widely spaced. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. In which situation do the particles of a substance have the highest average kinetic energy? 1

(a) ice at 0 degrees Celsius
(b) water at 25 degrees Celsius
(c) water at 90 degrees Celsius
(d) steam at 120 degrees Celsius

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. How can you separate a mixture of common salt and camphor? Which property is used? 2

12. How can the boiling point of water be raised and how can it be lowered? 2

13. Why does the temperature of boiling water not rise even when the flame is increased? 2

14. Explain why the particles in a liquid do not fly apart like those in a gas. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A glass of cold water on a table has water droplets on its outer surface. Explain the origin and why it appears more on a humid day. 3

16. Explain the process of how perfumes work using diffusion and evaporation. 3

17. Explain why it is easier to compress a gas than a liquid with the help of the data: air can be compressed to 1/10th, water practically not. 3

OR

Explain why spirit evaporates faster than water and cools more. 3

18. A mixture of two liquids is heated and one vaporises first. Name the concept and its use. 3

OR

Why does a camphor tablet get smaller on a hot day faster than on a cold day? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how steam burns, pressure cookers and boiling at hills are connected to latent heat and pressure. 5

OR

Two students release the smell of agarbatti at one end of a long hall at 10 degrees Celsius and at 35 degrees Celsius on two days. Compare the time for the smell to reach the other end and explain. 5

20. Discuss the role of evaporation in the water cycle and in regulating body temperature. Explain with particles. 5

OR

Describe how the properties of solid, liquid and gas decide their uses: giving two examples each. 5

END OF QUESTION PAPER — **Total: 40 Marks**